Token Management

Obtaining a Machine-Machine Token

Verifying a Token

Verifying a Token (Machine-Machine Calling)

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Obtaining a Machine-Machine Token

Function Description

This interface is used by a machine-machine user to apply for a token. A global machine-machine user can use the interface to apply for a token of other tenants. The validity period of tokens is 30 minutes (machine-to-machine calling).

Interface Protocol

HTTPS

Flow Control

50 times per second. If the number of requests exceeds the threshold, the system rejects the excess requests.

Request Method

POST

URI

/esf/iamservice/v1/token/machine-user/apply

Request

- **Message headers**

Parameter	Parameter Type	Mandatory	Value Range	Description

Parameter	Parameter Type	Mandatory	Value Range	Description
Content-Type	String	Yes	application/json	Set this parameter to **application/json**.

- **Message body**

```
{
"tenantId": "2000",
"accessKey": "2000|qF...h",
"secretKey": "58H...oR6R",
"machineUserAccount": "test123"
}
Or
{
"tenantAccount": "admin",
"accessKey": "2000|qF...h",
"secretKey": "58H...oR6R",
"machineUserAccount": "test123"
}
```

- **Parameters in the request body**

Parameter	Parameter Type	Mandatory	Value Range	Description
tenantId	String	No (mandatory if **tenantAccount** is empty)	None	Tenant ID.
tenantAccount	String	No (mandatory if **tenantId** is empty)	None	Tenant account (If the tenant ID is empty, use the tenant account to obtain the tenant ID to obtain the token.)
accessKey	String	Yes	None	AK information.
secretKey	String	Yes	None	SK information.
machineUserAccount	String	No	None	Machine-machine user account.

Response

- **Example**

```
{
"code": "1",
"msg": "success",
"data": {
"imoc_token": "gbT/ymOQQL81Jqz.............................",
"token_name": "imoc_token"
}
}
```

- **Parameters in the response body**

Field	Type	Description
code	String	Response code. The operations are as follows: **1**: success; other values: failure.
msg	String	Response message.
data	String	Response result.
imoc_token	String	Response token (EncodeURI required).
token_name	String	Response token name.

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Verifying a Token

Interface Function

Check whether the token is valid. (Human-to-machine calling)

Interface Protocol

HTTPS

Flow Control

50 times per second. If the number of requests exceeds the threshold, the system rejects the excess requests.

Request Method

POST

URI

/esf/iamservice/v1/auth/check/token

Request

- Message headers

  None

- Message body

Table 1 Parameters in the request body

Parameter	Value Type	Mandatory	Value Range	Description
token	String	Yes	None	**imoc_token** in cookies. Use the URLDecoder to decode.
updateTgt	Boolean	Yes	true/false	Renewal or not. The value can be **true** (yes) or **false** (no).

Example

- Message headers

  Content-Type:application/json

- Message body

```
{
      "token":"nlwt6t4UM5RiJ9",
       "updateTgt":false
}
```

Response

- Parameters in the response body

Parameter	Type	Value Range	Description
code	String	1/Other values	**1**: success; other values: failure
userId	String	None	Unique ID of a user
userAccount	String	None	User account
description	String	None	User description.
csrfToken	String	None	x-csrf-token
userName	String	None	User name
userType	String	None	**employee**: human-machine user; **machineUser**: machine-machine user
machineRoles	Array	None	Machine-machine role array
-id	String	None	Unique ID of a machine-machine role
-name	String	None	Machine-machine role name
tenantId	String	None	Unique ID of a tenant
roleId	String	None	Last used role ID

Parameter	Type	Value Range	Description
machineUser	String	None	Machine-machine user or not
ip	String	None	IP address of the accessed end.

- Example response for a successful request

```
{
    "code": "1",
    "msg": "Success",
    "data": {
        "userId": "1000",
        "userAccount": "admin",
        "csrfToken": null,
        "userName": "admin",
        "userType": "employee",
        "machineRoles": [{
             "id": "1001",   "name": "gam_authquery"
         }],
        "tenantId": "1000",
        "roleId": "1000",
        "machineUser": false,
        "ip": "10.146.4.90",
        "description": "master Administrator"
    }
}
```

- Example response for a failed request

```
{
    "code": "60200020002",
    "msg": "Failed to verify token",
    "data": null
}
```

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Verifying a Token (Machine-Machine Calling)

Function Description

This interface is used to check whether the token is valid.

Interface Protocol

HTTPS

Flow Control

50 times per second. If the number of requests exceeds the threshold, the system rejects the excess requests.

Calling Method

GET

URI

/esf/iamservice/v1/token/check

Request

- **Request example**
- Parameters in the request header

Parameter	Value Type	Mandatory	Value Range	Description
token	String	Yes	None	Machine-machine token.

- **Parameter description**

NOTE

Value types:

Number: numeric type.

String: string text value, for example, **INC-20200312-00000017**.

Boolean: **True** or **False**.

Array: [1,2,3,4] and ["value1","value2","value3"] are arrays. It is recommended that more than 2 values be placed in the example.

Object: [{"key": "exAttrInfo","value": "false"}, {"key": "exAttrInfo","value": "false"}] are objects. It is recommended that more than 2 values be placed in the example.

null: The object is null.

Response Message

- Parameters in the response body

Parameter	Data Type	Mandatory	Value Range	Description
iat	String	Yes	None	Time when the token takes effect.
exp	String	Yes	None	Expiration time of the token. ◦ The expiration time of the access token is 0.5 hour later than the effective time. ◦ The expiration time of the authorization token is that of the associated session.
userId	String	Yes	None	Unique identifier of a user.
userAccount	String	Yes	None	User account.
tenantId	String	Yes	None	Unique identifier of a tenant.
roles	String[]	Yes	None	◦ Role ID list of the machine-machine user is returned when the access token is parsed. ◦ No role information is returned when the

Parameter	Data Type	Mandatory	Value Range	Description
				authorization token is parsed.
iss	String	Yes	None	Token issuer. Currently, the value is fixed to **GAM**.
userType	Integer	Yes	None	User type.
sid	String	Yes	None	Login session ID.

- Example of a correct response packet

```
{
"resultCode": "0",
"resultMessage": "success",
"result": {
"iat": 1587679596480,
"exp": 1587765996480,
"userId": "100001",
"userAccount": "mu_gam",
"tenantId": "1000",
"roles": [{
"id": "1001",
"name": "gam_authquery"
},
{
"id": "1002",
"name": "gam_cas"
},
{
"id": "1003",
"name": "gam_masteradmin"
},
{
"id": "1004",
"name": "gam_tenantadmin"
}
],
"iss": "GAM",
"userType": "machineUser"
}
}
```

Status Codes

Response Code	Description
200 OK	Updated successfully – synchronous or query succeeded
400 Bad Request	The request is invalid.
401 Unauthorized	You do not have the permission to perform this operation.
403 Forbidden	The operation is rejected.
404 Not Found	The resource does not exist.
405 Method Not Allowed	The resource does not support the method.
500 Internal Host Error	The server does not execute the request.

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